Functional Annotation Report: *pdxA* (PP_0402, UniProt Q88QT5)

4-Hydroxythreonine-4-phosphate dehydrogenase — *Pseudomonas putida* KT2440

Summary

The gene **pdxA** (ordered locus **PP_0402**; UniProt **Q88QT5**) of *Pseudomonas putida* strain KT2440 encodes **4-hydroxythreonine-4-phosphate dehydrogenase** (**EC 1.1.1.262**), a cytoplasmic, divalent-metal-dependent, NAD(P)$^+$-linked oxidoreductase. Its primary and only known biochemical function is to catalyze the **fourth of five steps** of the **deoxyxylulose-5-phosphate (DXP)-dependent de novo pyridoxal 5′-phosphate (vitamin B6) biosynthesis pathway**. The enzyme performs an NAD(P)$^+$-dependent oxidation of **4-(phosphohydroxy)-L-threonine (HTP)** to an unstable β-keto-acid intermediate (2-amino-3-oxo-4-(phosphooxy)butyric acid) that spontaneously decarboxylates to yield the aminoketone **3-amino-2-oxopropyl phosphate** (also called 1-amino-3-hydroxyacetone phosphate, AHAP). This aminoketone is the direct substrate for the next pathway enzyme, PdxJ (pyridoxine-5′-phosphate synthase), which condenses it with deoxyxylulose-5-phosphate to build the pyridine ring of vitamin B6.

The functional assignment for PP_0402 rests on a combination of authoritative UniProt/HAMAP structured annotation and strong **evolutionary inference by orthology**. No experimental study of the *P. putida* PP_0402 protein specifically has been published; all direct biochemical and crystallographic evidence derives from the well-characterized *Escherichia coli* ortholog. A rigorous pairwise sequence analysis performed during this investigation established that Q88QT5 is **61.9 % identical (gap-free in the core)** to the *E. coli* PdxA, with **every catalytic and metal-binding residue conserved and colinear** — including the three Zn^{2+}-coordinating histidines and the two active-site-integrity aspartates (Asp247, Asp267). This provides high-confidence transfer of the *E. coli* functional and structural knowledge to the *P. putida* enzyme.

Structurally, PdxA is an **obligate homodimer** with a **shared, inter-subunit active site** located in a cleft at the dimer interface, contributed by residues from both monomers. Each active site binds one divalent metal ion (Zn^{2+}, or alternatively Mg^{2+} or Co^{2+}) coordinated by three conserved histidines. Despite low sequence similarity, PdxA adopts a fold structurally related to the **isocitrate dehydrogenase / isopropylmalate dehydrogenase (ICDH/IPMDH) superfamily** of β-decarboxylating dehydrogenases. Because the DXP-dependent B6 pathway is **absent in humans** (vitamin B6 is a required dietary nutrient for mammals), PdxA is expected to be **essential** for *P. putida* under conditions lacking exogenous B6, and represents a validated **selective antibacterial target class**.

Identity Verification

The mandatory verification steps required by the research brief were completed and all passed:

- **Gene symbol matches protein description.** The symbol *pdxA* is the canonical bacterial gene name for 4-hydroxythreonine-4-phosphate dehydrogenase, exactly matching the UniProt description for Q88QT5.
- **Organism confirmed.** The protein belongs to *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950), locus PP_0402, a 6.18 Mb metabolically versatile soil bacterium PMID: 12534463.
- **Family/domains align.** The protein carries the PdxA family signature (Pfam PF04166; InterPro IPR005255/IPR037510) and HAMAP-Rule MF_00536 — all consistent with the literature reviewed.
- **No cross-symbol ambiguity.** *pdxA* is not a widely reused symbol; the literature found describes bona fide PdxA enzymes. No conflicting gene of the same symbol in a different organism confounded the analysis.

The only caveat — flagged prominently below — is that **no study has characterized the PP_0402 protein directly**; the functional picture is transferred from the *E. coli* ortholog with which it shares 61.9 % identity and a fully conserved active site.

Key Findings

Finding 1 — PdxA is 4-hydroxythreonine-4-phosphate dehydrogenase (EC 1.1.1.262), the fourth step of DXP-dependent PLP biosynthesis

UniProt Q88QT5 annotates PP_0402 as 4-hydroxythreonine-4-phosphate dehydrogenase (EC 1.1.1.262) under HAMAP-Rule MF_00536, placing it firmly in the PdxA family (Pfam PF04166). The biochemical reaction is best defined through the extensively characterized *E. coli* ortholog. As described in the *E. coli* PdxA crystallography paper, *"the fourth step is catalyzed by 4-hydroxythreonine-4-phosphate dehydrogenase (PdxA, E.C. 1.1.1.262), which converts 4-hydroxy-L-threonine phosphate (HTP) to 3-amino-2-oxopropyl phosphate"* PMID: 12896974.

Two properties of the enzyme are especially diagnostic and are experimentally established in *E. coli*: **strict substrate specificity** for the phosphorylated form of the substrate, and **dual redox-cofactor flexibility**. The same study states that *"this divalent metal ion-dependent enzyme has a strict requirement for the phosphate ester form of the substrate HTP, but can utilize either $NADP^+$ or NAD^+ as redox cofactor"* PMID: 12896974. In other words, the free (non-phosphorylated) 4-hydroxythreonine is not a substrate — the enzyme reads out the phosphate ester — and the enzyme is promiscuous with respect to whether NAD^+ or $NADP^+$ serves as the hydride acceptor. This combination of tight substrate discrimination and cofactor flexibility is a hallmark of PdxA family enzymes.

Finding 2 — The exact reaction chemistry: an oxidative decarboxylation via a spontaneously-decarboxylating β-keto-acid intermediate

The structured UniProt annotation for Q88QT5 resolves the reaction to atomic precision. The FUNCTION field states that PdxA *"catalyzes the NAD(P)-dependent oxidation of 4-(phosphooxy)-L-threonine (HTP) into 2-amino-3-oxo-4-(phosphooxy)butyric acid which spontaneously decarboxylates to form 3-amino-2-oxopropyl phosphate (AHAP)"*. The canonical catalytic activity is registered as **RHEA:32275 / EC 1.1.1.262**:

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4-(phosphooxy)-L-threonine + NAD⁺  →  3-amino-2-oxopropyl phosphate + CO₂ + NADH
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Mechanistically, this is a **β-decarboxylating dehydrogenation**: the enzyme first oxidizes the C3 hydroxyl of HTP to a ketone (transferring a hydride to $NAD(P)^+$), generating an unstable β-keto acid (2-amino-3-oxo-4-(phosphooxy)butyric acid). This intermediate then loses CO_2 **spontaneously** (non-enzymatically), yielding the aminoketone product. This two-part chemistry — enzyme-catalyzed oxidation followed by spontaneous decarboxylation — mirrors the mechanistic logic of the ICDH/IPMDH superfamily to which PdxA is structurally related (see Finding 3).

The **cofactor** annotation specifies that the enzyme *binds 1 divalent metal cation per subunit* and can use **Zn^{2+}, Mg^{2+}, or Co^{2+}**. This metal flexibility is notable: while the crystal structure was solved with Zn^{2+}, the enzyme functions with alternative divalent cations, consistent with a catalytic/structural rather than strictly redox role for the metal. The **pathway** field places the reaction as *"pyridoxine 5′-phosphate biosynthesis; pyridoxine 5′-phosphate from D-erythrose 4-phosphate: step 4/5"*, and the **subcellular location** is annotated as **cytoplasm**.

Finding 3 — PdxA is a homodimeric, Zn^{2+}-containing enzyme with a shared inter-subunit active site, in the ICDH/IPMDH structural family

The *E. coli* PdxA crystal structure (solved with the HTP substrate and Zn^{2+} bound) reveals the enzyme's architecture. The protein *"forms tightly bound dimers"* PMID: 12896974, with each monomer adopting an α/β/α fold split into two subdomains. Critically, the **active site is located at the dimer interface**, within a cleft that receives contributions from both monomers — meaning the enzyme is an **obligate dimer** for catalysis, not merely for stability.

The metal cofactor sits at the heart of this shared site: *"a Zn^{2+} ion is bound within each active site, coordinated by three conserved histidine residues from both monomers"* PMID: 12896974. Two additional conserved residues stabilize the catalytic machinery: *"two conserved amino acids, Asp247 and Asp267, play a role in maintaining integrity of the active site"* PMID: 12896974.

Despite performing chemistry unique to the B6 pathway, PdxA is evolutionarily and structurally kin to a broad family of metabolic dehydrogenases: *"PdxA is structurally similar to, but limited in sequence similarity with isocitrate dehydrogenase and isopropylmalate dehydrogenase"* PMID: 12896974. ICDH and IPMDH are both **metal-dependent β-decarboxylating dehydrogenases**, so PdxA's placement in this superfamily is mechanistically coherent with the oxidative-decarboxylation chemistry described in Finding 2.

Finding 4 — *P. putida* PdxA (Q88QT5) is 61.9 % identical to the *E. coli* enzyme with all catalytic residues conserved and colinear

Because no PP_0402-specific experimental study exists, the reliability of transferring *E. coli* knowledge to *P. putida* hinges on sequence conservation. A global pairwise alignment performed in this investigation, between Q88QT5 (329 aa) and *E. coli* PdxA (P19624, 329 aa), yields **203/328 = 61.9 % identity**, essentially gap-free through the core. Both proteins are the same length, and the signature PdxA motifs are **perfectly colinear at identical positions**:

Motif	Approx. position	Role
`TPGEPAGIGPDL`	N-terminal	Rossmann-like NAD(P)-binding motif
`VHKGVIN`	~121	conserved core
`GHTEF`	~135	active-site region (His136)
`TTHLPLRD`	~164	active-site region (His166)
`NPHAGEGGH`	~209	diagnostic PdxA motif (Zn-coordinating His)

Crucially, the **metal- and substrate-binding residues characterized in the *E. coli* structure are all conserved** in Q88QT5. The three Zn^{2+}-coordinating histidines — the residues that the crystallography paper identifies as *"coordinated by three conserved histidine residues from both monomers"* PMID: 12896974 — are present at colinear positions (His→His at each). Likewise, the two active-site-integrity aspartates that *"play a role in maintaining integrity of the active site"* (Asp247, Asp267) PMID: 12896974 are conserved identically (Asp→Asp). This full conservation of the catalytic constellation is the strongest available evidence that PP_0402 is a **genuine, intact, functional PdxA** rather than a divergent paralog with an altered or lost function.

Finding 5 — PdxA feeds its aminoketone product to PdxJ in the DXP-dependent pathway, restricted to bacteria/fungi/plants

PdxA does not act in isolation; it is a committed enzyme feeding the ring-forming step. In *E. coli* and by orthology in gammaproteobacteria such as *P. putida* KT2440, de novo PLP *"is synthesized from the condensation of deoxyxylulose 5-phosphate and 4-phosphohydroxy-L-threonine catalysed by the concerted action of PdxA and PdxJ"* PMID: 17822383. PdxA generates the aminoketone (3-amino-2-oxopropyl phosphate), and PdxJ (pyridoxine-5′-phosphate synthase) condenses that aminoketone with DXP to form pyridoxine 5′-phosphate, which is subsequently oxidized to PLP by PdxH.

The enzymatic identities and sequential ordering are confirmed by a *Bacillus subtilis* engineering study, which lists *"the 4-hydroxy-L-threonine-phosphate dehydrogenase PdxA, the pyridoxine 5′-phosphate synthase PdxJ and the native DXP synthase, Dxs"* PMID: 25777134 as the reconstituted route — establishing PdxA and PdxJ as consecutive, cooperating enzymes. Importantly, this route is **taxonomically restricted**: *"its de novo biosynthesis occurs only in bacteria, fungi and plants, making it an essential nutrient in the human diet"* PMID: 17822383. The alternative single-enzyme Pdx1/Pdx2 (ribose-5-phosphate-dependent) route is used by most other organisms, but *P. putida* KT2440 encodes the DXP-dependent pathway.

Finding 6 — The pathway is absent in humans, making PdxA essential and a selective antibacterial target; no PP_0402-specific study exists

PLP is a cofactor of central importance: *"pyridoxal 5′-phosphate (PLP) is an essential cofactor for enzymes that catalyze diverse reactions in central metabolism"* PMID: 40439409 — including transaminases, decarboxylases, and racemases. Because the DXP-dependent de novo pathway *"occurs only in bacteria, fungi and plants"* PMID: 17822383 and is **absent in humans**, loss of PdxA is expected to render *P. putida* a pyridoxine auxotroph under B6-limited conditions, and the enzyme is therefore attractive as a **selective antibacterial target class**. A directed PubMed search during this investigation returned **no experimental study of the *P. putida* PP_0402 protein specifically**; all biochemical/structural characterization derives from the *E. coli* ortholog (with related metabolic engineering in *B. subtilis*).

Mechanistic Model / Interpretation

Position of PdxA in the DXP-dependent vitamin B6 pathway

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D-erythrose 4-phosphate                     pyruvate + glyceraldehyde-3-P
         │ (Epd)                                       │ (Dxs)
         ▼                                             ▼
4-phosphoerythronate                          1-deoxy-D-xylulose 5-P (DXP)
         │ (PdxB)                                      │
         ▼                                             │
(2-oxo-3-hydroxy-4-P-butanoate)                        │
         │ (SerC / PdxF)                               │
         ▼                                             │
4-(phosphohydroxy)-L-threonine (HTP)                   │
         │                                             │
         │   ┌───────────────────────────┐            │
         └──►│  PdxA  (PP_0402, Q88QT5)   │            │
             │  EC 1.1.1.262              │            │
             │  + NAD(P)⁺, divalent metal │            │
             └───────────────────────────┘            │
                         │                             │
             oxidation → β-keto acid                   │
             → spontaneous −CO₂                        │
                         ▼                             │
         3-amino-2-oxopropyl phosphate (AHAP) ────────┤
                                                       ▼
                                          ┌───────────────────────────┐
                                          │  PdxJ (PNP synthase)       │
                                          │  condenses AHAP + DXP      │
                                          └───────────────────────────┘
                                                       │
                                                       ▼
                                          pyridoxine 5′-phosphate (PNP)
                                                       │ (PdxH, O₂)
                                                       ▼
                                          pyridoxal 5′-phosphate (PLP)
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PdxA sits at the convergence of two branches of the pathway: one branch (via Epd → PdxB → SerC) generates HTP, PdxA's substrate; the second, independent branch (via Dxs) generates DXP. PdxA's product AHAP and DXP are the two building blocks that PdxJ stitches together into the pyridine ring. Thus PdxA is a **committed, non-redundant node**: its product is required stoichiometrically for ring formation, and there is no bypass within the DXP-dependent route. The upstream branch logic — that pdxA functions in a distinct branch from pdxB/serC — was established genetically in *E. coli* PMID: 2121717.

The catalytic cycle

1. **Substrate recognition.** HTP binds in the inter-subunit cleft; the enzyme's strict requirement for the **phosphate ester** form indicates specific recognition of the 4-phosphate group. The divalent metal (Zn^{2+}/Mg^{2+}/Co^{2+}), held by three conserved histidines, likely polarizes the C3 hydroxyl/carbonyl and stabilizes developing negative charge.
2. **Oxidation.** A hydride is transferred from C3 of HTP to NAD^+ or $NADP^+$, oxidizing the hydroxyl to a ketone and producing the β-keto acid 2-amino-3-oxo-4-(phosphooxy)butyric acid + NAD(P)H.
3. **Spontaneous decarboxylation.** The β-keto acid is chemically unstable and loses CO_2 without further enzyme catalysis, yielding **3-amino-2-oxopropyl phosphate (AHAP)**.
4. **Product release / hand-off.** AHAP is delivered to PdxJ for condensation with DXP.

This mechanism is a textbook example of the **β-decarboxylating dehydrogenase** logic shared with isocitrate dehydrogenase and isopropylmalate dehydrogenase — enzymes to which PdxA is structurally related despite low sequence identity. The metal flexibility (Zn/Mg/Co) and dual NAD^+/$NADP^+$ usage further echo this superfamily's characteristics.

Confidence in the annotation transfer

Evidence type	Source	Strength for PP_0402
EC number & reaction	UniProt/HAMAP MF_00536; RHEA:32275	High (curated)
Exact chemistry & intermediate	UniProt Q88QT5 structured annotation	High (curated)
Biochemistry (substrate specificity, cofactors, metal)	*E. coli* ortholog [PMID: 12896974]	High (transferred by 61.9 % identity)
Structure (dimer, shared site, Zn-His_3, Asp247/267)	*E. coli* crystal structure [PMID: 12896974]	High (all residues conserved & colinear)
Pathway context (PdxA→PdxJ)	[PMID: 17822383], [PMID: 25777134]	High
Direct PP_0402 experimental data	—	**None (gap)**

The overall picture is a **high-confidence functional annotation by orthology**, limited only by the absence of any organism-specific experimental validation for the *P. putida* protein itself.

Evidence Base

PMID	Title	Contribution
12896974	*Crystal structure of Escherichia coli PdxA, an enzyme involved in the pyridoxal phosphate biosynthesis pathway*	**Primary structural & biochemical reference.** Defines the reaction (HTP → 3-amino-2-oxopropyl phosphate), EC 1.1.1.262, strict phosphate-ester specificity, dual NAD^+/ $NADP^+$ use, homodimeric architecture, shared inter-subunit active site, Zn^{2+} coordination by three histidines, Asp247/Asp267 active-site integrity, and ICDH/IPMDH structural family membership. All key residues shown conserved in Q88QT5.
17822383	*Two independent routes of de novo vitamin B6 biosynthesis: not that different after all*	Places PdxA in the DXP-dependent pathway acting in *"concerted action"* with PdxJ; establishes taxonomic restriction (bacteria/fungi/plants only), underpinning essentiality and human-target selectivity.
25777134	*Engineering Bacillus subtilis for the conversion of the antimetabolite 4-hydroxy-L-threonine to pyridoxine*	Confirms enzymatic identities and sequential roles of PdxA (4-hydroxy-L-threonine-phosphate dehydrogenase) and PdxJ; demonstrates pathway reconstitution/engineering.
40439409	*Chemical proteomics enhances the understanding of 2AA stress*	Reinforces the central-metabolic essentiality of the PLP end-product PdxA helps synthesize.
2121717	*Metabolic relationships between pyridoxine and serine biosynthesis in E. coli K-12*	Provides upstream pathway context (SerC/PdxB branch generating HTP) and confirms pdxA functions in a distinct branch from pdxB/serC.
12534463	*Complete genome sequence... Pseudomonas putida KT2440*	Establishes the genomic/organismal context (6.18 Mb genome, metabolically versatile biosafety strain) in which PP_0402 resides.
26913973	*The revisited genome of Pseudomonas putida KT2440...*	Updated functional annotation of the KT2440 chromosome (re-annotation of ~1548 genes), the reference framework for PP_0402.

Several additional papers surfaced during literature searches (e.g., alanine racemase, L-methioninase, PLP-dependent probe studies) are relevant only to the **downstream utilization of PLP** by other enzymes, not to PdxA itself; they contextualize why the PdxA product matters but do not directly characterize PdxA.

Limitations and Knowledge Gaps

1. **No organism-specific experimental data.** The single most important limitation is that **no biochemical, structural, genetic, or physiological study of the *P. putida* PP_0402 protein has been published.** The functional annotation is entirely inferred — from curated UniProt/HAMAP rules and by orthology to the *E. coli* enzyme. While the 61.9 % identity and full active-site conservation make this inference robust, kinetic parameters (K_m for HTP, NAD^+ vs $NADP^+$ preference, metal preference, kcat) for the *P. putida* enzyme are unmeasured.
2. **Essentiality is inferred, not demonstrated in *P. putida*.** The prediction that PP_0402 is essential (under B6-limited conditions) rests on pathway logic and the human-absence of the route, not on a *P. putida* knockout or transposon-insertion phenotype. Genome-wide essentiality datasets for KT2440 were not directly interrogated in this investigation.
3. **Metal preference in vivo is uncertain.** UniProt lists Zn^{2+}/Mg^{2+}/Co^{2+} as possible cofactors, and the *E. coli* crystal structure used Zn^{2+}. Which metal is physiologically employed by the *P. putida* enzyme, and whether metal identity modulates activity, is unresolved.
4. **Substrate hand-off to PdxJ is assumed, not shown.** Whether AHAP is channeled to PdxJ via a physical complex or released into the bulk cytoplasm is not established for *P. putida* (or definitively for *E. coli*).
5. **Regulation and expression context unknown.** The operon structure, transcriptional regulation, and expression level of pdxA/PP_0402 in *P. putida* (and any response to B6 availability) were not examined.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and enzyme kinetics.** Clone PP_0402, express and purify the recombinant protein, and measure kinetic constants (K_m, kcat) for HTP with both NAD^+ and $NADP^+$, and across Zn^{2+}/Mg^{2+}/Co^{2+}, to confirm the transferred biochemistry and quantify the *P. putida*-specific parameters.
2. **Functional complementation.** Test whether PP_0402 complements an *E. coli* ΔpdxA pyridoxine auxotroph, providing a direct genetic demonstration that PP_0402 encodes a functional 4-hydroxythreonine-4-phosphate dehydrogenase.
3. **Targeted deletion / conditional knockout in *P. putida* KT2440.** Construct a ΔPP_0402 strain and assay for pyridoxine/pyridoxal auxotrophy (growth rescued by exogenous B6). This would experimentally confirm both function and essentiality within the native organism.
4. **Structure determination.** Solve the crystal or cryo-EM structure of the *P. putida* enzyme (ideally with HTP and metal bound), or generate a high-confidence AlphaFold model, to verify the predicted dimer interface, shared active site, Zn-His_3 coordination, and Asp247/Asp267 positioning.
5. **Metabolic/pathway mapping.** Confirm co-transcription/co-regulation with neighboring pdx/ser genes and probe substrate channeling to PdxJ (e.g., via co-purification or activity coupling assays).
6. **Antibacterial target validation.** Given the human-absence of the pathway, evaluate PP_0402 (and its orthologs in pathogenic *Pseudomonas*) as a target for selective inhibitor development, building on the ICDH/IPMDH-related active-site architecture.

Conclusion

pdxA (PP_0402, Q88QT5) in *Pseudomonas putida* KT2440 encodes **4-hydroxythreonine-4-phosphate dehydrogenase (EC 1.1.1.262)**, a cytoplasmic, homodimeric, divalent-metal-dependent, $NAD(P)^+$-linked oxidoreductase that catalyzes the **fourth step of the DXP-dependent de novo pyridoxal 5′-phosphate (vitamin B6) biosynthesis pathway**. It oxidizes 4-(phosphohydroxy)-L-threonine to an unstable β-keto acid that spontaneously decarboxylates to the aminoketone 3-amino-2-oxopropyl phosphate, which PdxJ then condenses with deoxyxylulose-5-phosphate to form the vitamin B6 ring. The assignment is supported by curated database annotation and by high-confidence orthology (61.9 % identity, fully conserved active site) to the crystallographically and biochemically characterized *E. coli* enzyme, though no experimental study of the *P. putida* protein itself yet exists.
